

Proforma for Submitting Technical Proposal

1. General Information

S. No.	Particulars	Details to be Filled by Team
1	Hackathon Title	C-DAC SUVIDHA Challenge 2026
2	Problem Statement ID / Theme	Unified Civic Utility Kiosk Interface
3	Team Name	Arditechs
4	Institution / Organisation Name	NIT SILCHAR
5	Team Leader Name	Anubrata Paul
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8	City / State	Assam

2. Team Composition

S. No.	Name of Team Member	Role in Team	Qualification / Specialization
1	Anubrata Paul	API, Security and Database	B.Tech 1 st year Student
2	Sakshi Sharma	Backend Developer	B.Tech 1 st year Student
3	Kislay Kumar	UI/UX Designer	B.Tech 1 st year Student
4	Aman Sagar	Frontend Developer	B.Tech 1 st year Student

3. Project Overview

Project Title:

SUVIDHA: An Integrated Multi-Modal Civic Utility Kiosk with Stateless Security Architecture

Brief Project Description:

SUVIDHA is a high-availability kiosk solution designed to decentralize and simplify essential civic services for the "next billion" users. By utilizing a Unified Identity Resolution Layer, the platform consolidates fragmented utility services—including Electricity, Gas, and Municipal Grievance Redressal—into a single, touch-optimized interface. The system bridges the digital divide by offering a low-friction entry point for

citizens who lack personal computing devices or high technical literacy, ensuring government services are accessible 24/7 in a secure public environment.

Key Features & Functionalities:

Key Features & Functionalities: Unified Multi-Parameter Authentication: A proprietary backend logic that resolves identity across three distinct identifiers (Aadhaar, Registered Mobile, or Consumer ID) without requiring user-side selection, drastically reducing "Form Friction."

- **Privacy-First Data Vault:** Implementation of "Privacy by Design" via Masked Aadhaar Storage and terminal-end data purging, ensuring full compliance with the DPDP Act 2023.
- **Asynchronous Micro-Service Bridge:** A Node.js/Express backend that decouples different utility providers, ensuring that a downtime in one service (e.g., Gas) does not affect the availability of others (e.g., Electricity).
- **Localized Multilingual Interface:** Native support for regional languages, specifically optimized for the diverse linguistic landscape of Assam and the Northeast, ensuring inclusive governance.
- **Stateless Session Management:** Utilization of JWT (JSON Web Tokens) to ensure that no citizen data is cached or persisted on the physical kiosk hardware after a session concludes
- **User Friendly Chat Bot :** "Sahayak" AI Assistant our multilingual chatbot will act as an active assistant available 24/7 which provides step-by-step instructions, multilingual support, and visual prompts to ensure "easy navigation" for diverse customer groups

4. Problem Understanding & Objective

4.1 Understanding of the Problem Statement

The current landscape of civic service delivery in India faces a "Fragmentation Crisis." Despite the digital revolution, three primary barriers prevent the "Last Mile" citizen from accessing essential services:

1. **Digital Fragmentation:** Services such as APDCL (Electricity), Gas (Pradhan Mantri Ujjwala Yojana), and Municipal grievances are hosted on disparate, siloed portals. Each requires unique login credentials and follows different UI/UX patterns, leading to cognitive overload for the user.
2. **The Identity Bottleneck:** Most portals force a "top-down" authentication approach where a user must remember exactly which ID (Consumer No. or Phone or Aadhaar) is linked to which service. This creates a high barrier to entry for the non-tech-savvy population.
3. **Hardware & Literacy Gap:** A significant portion of the population in regions like the Northeast lacks personal high-speed internet or high-end smartphones. Public Service Centers (CSCs) are often overcrowded or limited by human operating hours.

Our understanding is that there is an urgent need for a Physical-Digital (Phyigital) Bridge—a kiosk that is as easy to use as an ATM but as secure as a bank vault.

4.2 Objectives of the Proposed Solution

The primary objective of the SUVIDHA project is to engineer a resilient, user-centric gateway for public services with the following specific goals:

- **Unified Service Convergence:** To integrate multiple utility APIs into a single, cohesive dashboard, allowing a citizen to pay an electricity bill and file a municipal complaint in under two minutes.
- **Zero-Friction Authentication:** To implement an Identity Resolution Engine that allows users to authenticate using any available identifier (Aadhaar/Mobile/Consumer ID), removing the need for

pre-selection and reducing entry errors.

- **Privacy-by-Design Architecture:** To establish a security framework that adheres to the DPDP Act 2023, ensuring that sensitive data is masked and sessions are stateless, thereby protecting citizens in a public environment.
- **High Availability & Scalability:** To build the backend using Node.js and MySQL with a microservices approach, ensuring the kiosk can scale to handle millions of transactions across a Smart City network without performance degradation.
- **Linguistic Inclusivity:** To deploy a multilingual interface (specifically focusing on Assamese and English) to ensure the solution is inclusive of all demographic segments.

5. Technical Design & Architecture

5.1 Solution Description

The proposed solution is a Unified Citizen Service Kiosk Network designed to improve public service accessibility and digital governance across the city. The kiosks will be strategically installed at high-footfall locations such as railway stations, bus terminals, airports, government offices, and major marketplaces. The objective is to create a single, convenient touchpoint where citizens can access multiple civic and utility services without needing a personal smartphone, computer, or reliable internet connection.

Through this kiosk, a citizen can perform several essential tasks, including registering complaints, paying utility bills, submitting service applications, tracking requests, and obtaining information regarding public schemes. The platform integrates services from various departments such as the Municipal Corporation, Electricity Board, Water Supply Department, Gas Agency, and other civic authorities. Instead of visiting multiple offices, waiting in queues, or navigating complicated online portals, users can complete the entire process within minutes at a nearby kiosk.

When a citizen approaches the kiosk, the interface guides them step-by-step using clear prompts and visual instructions. After selecting the required service, the user enters basic details and submits the request. The system automatically routes the complaint or application to the appropriate department. Each request is assigned a tracking ID, allowing users to monitor progress and receive updates.

A major focus of the solution is security, reliability, and inclusivity. The kiosk is designed to be safe for sensitive public use and accessible for people of all ages and literacy levels. To achieve this, the system incorporates several advanced core features:

1. **Privacy-First Data Vault:** The platform follows a strict Privacy by Design principle. Aadhaar or identity information is stored only in masked form, and all personal data is automatically purged from the kiosk terminal after the session ends. This ensures compliance with the Digital Personal Data Protection (DPDP) Act, 2023, protecting citizens from data misuse or identity theft.
2. **Stateless Session Management:** The kiosk uses JWT (JSON Web Tokens) authentication. No citizen data is cached or permanently stored on the physical hardware. Once a user logs out or the session expires, the system clears all information, making the kiosk safe even in public environments.
3. **Asynchronous Micro-Service Bridge:** A Node.js and Express-based backend architecture connects multiple government and utility providers through independent micro-services. This decoupled system

ensures reliability—if one service (for example, gas supply) is temporarily unavailable, other services (such as electricity bill payment or complaint registration) continue functioning without interruption.

4. **Localized Multilingual Interface:** The kiosk supports multiple regional languages, especially those widely spoken across Assam and the Northeast. Citizens can interact with the system in their preferred language, making the service accessible to rural users, elderly individuals, and non-English speakers, thereby promoting inclusive governance.

5. **“Sahayak” AI Assistant** An integrated multilingual chatbot, Sahayak, acts as a 24/7 virtual assistant. It provides voice and text guidance, step-by-step instructions, and visual prompts to help users navigate the kiosk. This feature is particularly useful for first-time users and digitally inexperienced citizens.

Overall, the unified kiosk system reduces administrative burden, shortens service delivery time, and enhances transparency. It bridges the digital divide by ensuring that government services are available to every citizen—efficiently, securely, and conveniently

5.2 System Architecture Diagram

https://www.canva.com/design/DAHA7ownxtg/KE_XtpnCkm033ZPU6mKTDA/edit?utm_content=DAHA7ownxtg&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

5.3 Workflow & Data Flow

The system follows a structured, unidirectional data flow designed for security and speed. The workflow is divided into three primary layers: the User Interaction Layer, the Processing Layer (API), and the Data Persistence Layer (Database).

1. Authentication Workflow

- **Input:** The citizen provides an identifier (Consumer ID / Mobile / Aadhaar Suffix).
- **Request:** The Kiosk frontend sends a POST request to the /api/auth/unified-login endpoint.
- **Verification:** The Node.js server executes an indexed SQL query to verify the identity. If a match is found, a simulated OTP challenge is triggered.
- **Tokenization:** Upon OTP verification, the server generates a JWT (JSON Web Token). This token is sent back to the Kiosk to authorize all future requests during that session.

2. Service Selection & Data Fetching Workflow

- **Selection:** The user selects a specific service (e.g., "Electricity Bill Payment" or "Property Tax").
- **Authorization:** The Kiosk sends the request along with the JWT in the header. The API middleware verifies the token's validity.
- **Data Retrieval:** The API bridge queries the respective MySQL utility tables. The data (Amount Due, Due Date, etc.) is returned as a JSON object and rendered on the Kiosk UI.

3. Transaction & Update Workflow (Data Flow)

- **Action:** The user initiates a payment or submits a service request (e.g., LPG Refill).
- **Processing:** The API processes the request. For payments, it interacts with a secure payment gateway.
- **Commit:** Once successful, the API executes an ACID-compliant SQL update. The utility_bills table status changes to 'Paid', and a new entry is created in the security_logs table for auditing.

- **Confirmation:** The final response is sent to the Kiosk to display a success message, and a notification packet is sent to the user's registered mobile via an SMS gateway.

4. Security Data Flow Highlights

- **Encryption:** Data in transit is protected via TLS 1.3.
- **Masking:** Sensitive data like Aadhaar is masked at the database level and never travels across the network in full form.
- **Statelessness:** No citizen data is "stored" at the Kiosk layer; the data flow is strictly temporary and cleared immediately after the session expires or the user logs out.

6. Technical Details

1. Tools & Frameworks Used:

○ Frontend:

- **React.js (v18+):** Leveraged for its component-based architecture to build a fast, modular User Interface. This allows for high-speed screen transitions, which is critical for reducing citizen wait times at the kiosk.
- **Tailwind CSS:** Utilized to implement a mobile-first, responsive design. It ensures the kiosk UI is accessible, featuring high-contrast themes and large, touch-friendly hit areas for elderly citizens and those with motor impairments.
- **Framer Motion:** Used for subtle visual cues and animations that guide the user through the multi-step bill payment process, reducing "cognitive load" for non-tech-savvy users.

○ Backend:

- **JWT (JSON Web Tokens):** Implemented for stateless session management. Upon authentication, the server issues a signed token, ensuring that sensitive citizen data is never stored in the kiosk's local storage or in unencrypted cookies.
- **Bcrypt.js:** Employed for military-grade hashing of administrative credentials and sensitive PINs, ensuring that even in the event of a database leak, user data remains unreadable.
- **Node.js & Express.js:** An asynchronous, event-driven runtime chosen for its ability to handle thousands of concurrent I/O operations. This ensures the kiosk remains responsive even when multiple units are fetching utility data from the server simultaneously.

○ Database:

- **MySQL 8.0:** Chosen for its proven reliability and strict ACID (Atomicity, Consistency, Isolation, Durability) compliance. This is critical for financial transactions (bill payments) where data integrity is non-negotiable.
- **Multi-Key Indexing:** We have optimized the citizens table with B-Tree indexes on consumer_id, phone_number, and masked_aadhaar to ensure the Unified Identity Resolver fetches records in sub-100ms.
- **Connection Pooling:** Utilized to manage database connections efficiently, preventing server crashes during peak usage hours in municipal offices.

○ APIs:

- **RESTful API Architecture:** A clean, resource-oriented API design that decouples the frontend from the backend, allowing for independent updates to each layer.
- **Unified Auth Resolver:** A custom internal API logic that accepts a single input and intelligently routes the search across multiple identity columns.
- **Integration Ready:** Built with future-ready hooks to integrate with the Bharat Bill Payment System (BBPS) and Aadhaar Auth (UIDAI) via secure HTTPS gateways.

1. Hardware Components (if applicable):

The hardware architecture of the **CivicGuard SUVIDHA Kiosk** is designed for extreme durability, high-security data processing, and inclusive accessibility. It features a ruggedized edge-computing setup with multiple redundancy layers for connectivity.

1. Core Processing & Logic Unit

- **Industrial Mini-PC / Raspberry Pi 4:** Serves as the localized server hosting the Node.js runtime. It manages the Unified Identity Resolver and local encryption (AES-256) of data before transmission.

2. User Interaction & Data Capture

- **15.6" Capacitive Touchscreen:** A high-brightness, vandal-resistant panel designed for multi-touch interaction with the React.js frontend.
- **Integrated A4 Document Scanner:** A specialized CIS-based scanner used to digitize handwritten complaints, written grievances, or physical documentation (ID proofs) for municipal records.
- **QR/Barcode Scanner:** Facilitates rapid bill fetching from physical paper bills and supports UPI-based QR payments.

3. Secure Output & Transaction Hardware

- **Encrypted Thermal Printer:** Provides instant, tamper-evident physical receipts for all financial transactions, including Electricity, Gas, and Water bill payments.

4. Hybrid Communication System (Northeast Resilience)

- **Satellite Transceiver Module (VSAT/Starlink):** Acts as the critical backhaul for remote "Shadow Zones." It ensures the kiosk remains functional in hill districts where cellular networks are unreliable.
- **4G/LTE Failover Module:** Provides secondary terrestrial connectivity, automatically switching between cellular and satellite paths to ensure 100% uptime for the MySQL database sync.

5. Resilience & Power

- **Internal UPS:** Protects SQL data integrity by providing a power window to complete active transactions during the frequent power fluctuations common in the region.

7. Implementation Details

How does the solution work?

The **CivicGuard Smart City Hub** operates as a centralized "Service Aggregator." The implementation follows a structured 6-step workflow to ensure high accessibility and government-grade security:

1. **Unified Authentication Layer:** The citizen interacts with the kiosk's touch interface. Instead of navigating separate login pages for different departments, the user enters a single identifier—either their Registered Mobile Number, Consumer ID, or the last 4 digits of their Aadhaar. The Node.js backend performs a multi-key resolution against the MySQL database to identify the citizen.
2. **Identity Verification (Two-Factor Authentication):** Once the account is identified, the system triggers a simulated OTP sent to the registered mobile number. This provides a second layer of security, ensuring that sensitive utility data and payment histories are protected against unauthorized access.
3. **Dynamic Service Dashboard:** Upon successful login, the citizen is presented with a modular dashboard categorized by department. Each department offers specific digital services:
 - **Electricity:** View/Pay Bills, Request Load Extension, and Power Fault Reporting.
 - **Gas:** LPG Refill Booking, New Connection (Ujjwala Yojana) Registration, and Safety Audit booking.
 - **Municipal:** Property Tax payment, Birth/Death Certificate requests, and Public Grievance Redressal (e.g., streetlights, waste collection).
4. **Secure Data Fetching via API Bridge:** When a service is selected (e.g., "Property Tax"), the Node.js API fetches real-time records from the specific department's table in MySQL. The system utilizes JSON Web Tokens (JWT) for session management, ensuring that all data in transit is encrypted and the session is stateless—meaning no citizen data is stored on the physical kiosk hardware.
5. **Request Processing & Transaction:** For payments, the system integrates with a secure payment gateway (simulated for prototype). For non-payment services, like a "New Connection Request," the system generates a unique Service Tracking ID, logs the request in the security_logs table, and updates the departmental database status to "Pending."
6. **Instant Confirmation & Digital Receipting:** Once the transaction or request is completed, the MySQL database is updated using ACID-compliant transactions to prevent data corruption. A visual receipt is displayed on the kiosk, and a confirmation SMS containing the digital receipt or tracking ID is instantly pushed to the citizen's mobile device.
7. **Smart Assistance :** Throughout the process, the user can activate the "Sahayak" AI Assistant via voice or text for multilingual guidance in local languages like Assamese or Hindi.

8. Impact & Future Scope

8.1 Innovation & Uniqueness

The CivicGuard SUVIDHA interface distinguishes itself from existing government portals through several "first-of-its-kind" technical implementations:

- **Unified Identity Resolution Engine:** Unlike standard interfaces that require pre-selection of login methods, our "Single-Input Multi-Match" algorithm resolves identity across Aadhaar suffixes, mobile numbers, or Consumer IDs in real-time. This eliminates "Form Friction" and simplifies the "Citizen Interaction" experience.
- **"Sahayak" Virtual Interactive Assistant:** We have integrated a "Smart Urban Virtual Interactive" layer that provides "visual prompts and step-by-step instructions." This ensures that the platform is accessible to citizens regardless of their technical literacy.
- **Decoupled Micro-Service Bridge:** The backend utilizes an "independent, loosely coupled" architecture. This ensures that a localized failure in one utility provider's API (e.g., Gas) does not disrupt the "centralized access" to other critical services like Electricity or Grievance tracking.
- **Privacy-by-Design Compliance:** The architecture inherently adheres to the DPDP Act 2023 by utilizing "Data Masking" and "Stateless Session Management (JWT)," ensuring that sensitive "Aadhaar or identity information" is never persisted on public kiosk hardware.

8.2 Future Improvements & Scalability:

The project is designed to be a "scalable and convenient" foundation for a city-wide "Unified Citizen Service Kiosk Network":

- **Biometric & Edge-AI Integration:** Future iterations will incorporate "Physical Biometric Integration" (Fingerprint/Iris) and "Edge-AI" based Face Recognition for faster, password-less "secure citizen authentication."
- **Conversational AI Evolution:** The "Sahayak" assistant will evolve into a full "Conversational AI Entity," utilizing "Natural Language Processing (NLP)" to handle complex service requests via voice in localized regional dialects.
- **IoT Smart City Hub:** We plan to integrate the kiosk with "CivicGuard Smart City" sensors to provide "real-time emergency information display," such as weather alerts, air quality monitoring, and disaster notifications.

9. Scalability & Extensibility

I. System Scalability

Our architecture is designed to scale from a single prototype to a city-wide "Unified Citizen Service Kiosk Network" without performance bottlenecks:

- **Horizontal Scalability (Micro-Service Architecture):** By utilizing Node.js and Express, the backend is designed as an "Asynchronous Bridge." This means we can add new government departments (e.g., Transport or Health) as independent micro-services without rewriting the core identity resolver.
- **Database Partitioning & Indexing:** The MySQL 8.0 database uses B-Tree Indexing and is ready for "Vertical Partitioning." This allows the system to manage millions of citizen records and "Consumption Data" across multiple districts while maintaining sub-second query speeds.
- **Cloud-Ready Deployment:** The entire backend is containerized (Docker-ready), allowing it to be deployed on C-DAC's Meghraj Cloud or any centralized data center. This ensures the kiosk network can handle "High-Concurrency" during peak bill-payment cycles (e.g., month-end).

II. Universal Accessibility

In line with the "SUVIDHA" mission of bridging the digital divide, the interface ensures "Ease of navigation" for all demographic segments:

- **"Sahayak" AI-Driven Guidance:** The integrated Sahayak Assistant provides "Step-by-step instructions and visual prompts," specifically assisting citizens with low digital literacy or those who cannot navigate complex menus.
- **Localized Multilingual Support:** The interface is built with a "Multilingual UI," offering full support for **Assamese, Bodo, Hindi, and English**. This ensures that rural and elderly users can access "Essential civic services" in their native tongue.
- **Adaptive UX for Inclusive Use:**
 - * **Visual Accessibility:** High-contrast modes and "Voice-guided navigation" for visually impaired citizens.
 - **Physical Accessibility:** The touch-interface is designed with large hit-zones for users with limited motor skills.
- **Low-Bandwidth Optimization:** The "Loosely coupled" data flow is optimized for low-data consumption, ensuring the kiosk remains "Reliable and convenient" even in areas with inconsistent internet connectivity.

10. Security & Compliance

Aspect	Description
User Authentication	Multi-Factor Unified Identity Resolver: Authentication is handled through a two-step process. First, identity is resolved using indexed lookups for Consumer ID, Mobile, or Aadhaar suffix. Second, a time-sensitive TOTP (Time-based One-Time Password) is triggered to the registered mobile number, ensuring high-assurance identity verification.
Data Privacy	DPDP-Compliant Data Minimization: Following the "Privacy by Design" principle, the system avoids storing PII (Personally Identifiable Information) in plain text. Aadhaar Vaulting techniques are used to store only the terminal 4 digits, ensuring that the full 12-digit UIDAI number never resides on the application server or kiosk hardware.
Compliance with Govt. IT / DPDP Act	IT Act & DPDP Act (2023) Alignment: The solution adheres to the Digital Personal Data Protection Act regarding "Data Principal" rights. It includes automated logs for data processing and follows the CERT-In guidelines for secure application development and incident reporting.
Secure Transactions	Stateless JWT & ACID Integrity: All financial interactions are secured using JSON Web Tokens (JWT) with a 15-minute expiration window to prevent session hijacking on public kiosks. Database updates for bill payments are wrapped in ACID-compliant transactions to ensure zero data corruption during network fluctuations.

11. Declaration

We hereby declare that the information provided in this technical proposal is true and correct to the best of our knowledge. The proposed solution is our original work and does not violate any intellectual property rights.

Anubrata Paul

Signature: